Monopoly Screening of Environmentally Conscious Consumers with Carbon Offsets

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Abstract

In this paper, I characterize market equilibria if the seller has access to an offsetting technology to mitigate the pollution that is generated by the consumption and production of the core product it sells. In contrast to an intrinsic environmental quality, offsets do not tie the environmental quality to the core product sold to a consumer, but they also allow consumers to pay to reduce the pollution emitted by the core product sold to other consumers. I show that whether and how a monopolistic seller employs these offsets in equilibrium depends on the nature of the environmental preferences of conscious consumers. If consumers have responsibility-based preferences and care exclusively about the net pollution of their own consumption good, the seller maintains high demand and profits by offering no offsets to selfish consumers and full offsets for the pollution of the core product sold to environmentally concerned consumers. Because the pollution emitted by the core product sold to selfish consumers is not mitigated, this market outcome is inefficient, and a benevolent social planner wants to correct this inefficiency by setting a carbon tax, which works as a mandatory offset for all core products. In contrast, if consumers have outcome-based preferences and care about the total pollution emitted in the market, a seller benefits by expanding the product line and offering the most environmentally concerned consumers the possibility to pay to offset the pollution emitted by the goods sold to selfish consumers. Under certain conditions, the aggregate amount of pollution can be offset. Work in progress extends the results to a duopolistic setting.

1 Introduction

How can private firms and consumers contribute to the fight against global warming? Although economic theory has long advocated market-based policies to mitigate carbon emissions from the production and consumption of products, the implementation of carbon taxes (Pigou, 1920) or cap-and-trade mechanisms (Dales, 2002) remains slow (Bénabou and Tirole, 2010; Aghion et al., 2023). As of 2021, explicit carbon taxes exist in only 30 jurisdictions worldwide (Metcalf, 2021). Willing to incur private costs to reduce environmental harm, environmentally conscious consumers have the potential to fill the action gap left by lagging governance. On the supply side, sellers can exploit consumers' intrinsic motivation and benefit from vertically differentiating their products and segmenting the market along environmental quality dimensions (Mussa and Rosen, 1978). However, less attention has been given to how the integration of voluntary carbon offsets shapes product design and market efficiency. The voluntary carbon market is projected to reach $50 billion by 2030 (Blaufelder et al., 2021), with firms increasingly embedding offsets directly into

product offerings—such as in aviation (Berger et al., 2022) and online retail (Rodemeier, 2023). How the availability of voluntary carbon offsets affects market outcomes depends crucially on how consumers perceive their responsibility for pollution. Consumers may care strictly about the emissions tied directly to their own consumption (Guias, 2026), or they may care about their environmental impact regardless of who generated the emissions (Fehr and Schmidt, 1999; Charness and Rabin, 2002).

In this paper, I study the optimal product design of sellers who want to sell a core product and have access to voluntary carbon offsets. The difference to an intrinsic environmental quality is that voluntary carbon offsets can be sold to a consumer but reduce the pollution emitted by the core product sold to other consumers. Specifically, I study how optimal product-offset design is shaped by environmentally conscious consumers who care about the total pollution in the market, thus having outcome-based preferences (Fehr and Schmidt, 1999; Charness and Rabin, 2002), as opposed to consumers who care about the pollution of their own consumption good, thus having responsibility-based preferences. If the source of consumers' intrinsic motivation to reduce pollution stems from responsibility-based preferences, I find that the seller maintains high demand by offering full carbon offsets to *all* environmentally concerned consumers but not to purely selfish consumers. Because the pollution caused by selfish consumers is not compensated, this market is inefficient and I show that a social planner wants to correct the inefficiency with a carbon tax. However, the optimal product design changes if the source of intrinsic motivation to reduce pollution is outcome-based. Then the seller optimally offers a three-tiered menu: purely selfish consumers do not receive any carbon offsets, consumers with low-to-moderate environmental concerns receive full offsets for their product, and consumers with high environmental concerns receive offsets that compensate for the pollution of their own product *and* of the products sold to selfish consumers. Because the entire pollution is compensated, this market is efficient.

To answer this research question, I study a monopolistic seller who wants to sell a core product and faces consumers who value the intrinsic quality of the core product and differ in their valuation of the pollution emitted by the product's production. The seller does not observe consumer preferences, but knows the distribution of consumers' valuations of pollution and whether consumers care exclusively about the pollution tied to their product or about the overall environmental impact of their consumption. The seller has access to an offsetting technology, which allows bundling consumption products with voluntary carbon offsets that reduce pollution by a specified amount at a convex quadratic offsetting cost. Crucially, the amount of voluntary carbon offsets per product is not tied strictly to the pollution emitted by an individual product, but is capped by aggregate pollution emitted in the market. Furthermore, the seller knows and takes the cost parameters of employing voluntary carbon offsets as given.

The employment of voluntary carbon offsets creates incentives for the seller to partition the market into two segments by offering distinct product-offset contracts. The first segment consists of all consumers who show no concern for the pollution emitted by the production of consumption products. I show that in any equilibrium it must be true that this market segment receives the core product without any offsets, regardless of the source of conscious consumers' intrinsic motivation to reduce pollution. The monopolistic seller extracts all their surplus for the core product. Offering offsets is costly for the seller and selfish consumers are not willing to pay for them. This decision does not affect the demand of environmentally conscious consumers—either because they do not care about the pollution generated by products of other consumers or because they cannot influence the seller's choice to offer no offsets to purely selfish consumers.

Furthermore, offering carbon offsets is incentive-compatible only if they offset at least the pollution caused by the production of the product because a conscious consumer's willingness to pay for a core product with partial offsets is strictly below that of a selfish consumer. If a seller offered a menu option with partial offsets, purely selfish consumers would select into that option.

The menu offered to conscious consumers depends on the structure of their environmental preferences. If conscious consumers care exclusively about the emissions tied to their product, they are not willing to pay for additional offsets beyond their product's carbon footprint. Therefore, the seller offers a two-option menu in equilibrium: a core product without offsets to selfish consumers and a product with offsets that fully compensate the product's emissions to conscious consumers. If conscious consumers care instead about their overall environmental impact, then the seller adds a third option to the product line: high-conscious types are offered the core product with offsets that overcompensate their product's carbon emissions. By offering overcompensation, the seller extracts high-conscious types' willingness to pay to compensate for the pollution emitted by the production of products sold to selfish consumers.

Turning to the policy side, I show that a benevolent social planner who cares about consumer surplus, profits, and total pollution emitted in the market wants to offset aggregate pollution emitted in the market. Hence, the market outcome is inefficient when consumers exclusively care about the pollution emitted by their own product. When allowed to set a carbon tax, the social planner corrects for this inefficiency by setting it equal to the marginal social cost of pollution. In further research, I plan to extend the analysis to a duopolistic setting, studying how competition affects market outcomes when sellers compete with product lines consisting of a core product and carbon offsets.

This paper makes two key contributions: first, it shows that the nature of environmental preferences determines whether markets achieve social efficiency when using voluntary carbon offsets. Second, it provides a novel mechanism showing how voluntary carbon offsets can be leveraged by firms to internalize third-party externalities when consumers hold outcome-based environmental preferences. This paper proceeds as follows: section 2 presents the model set-up. Section 3 discusses the resulting market equilibria if environmentally conscious consumers have responsibility-based and outcome-based preferences. Section 4, which is work in progress, discusses welfare implications of the two resulting market equilibria. Section 5 concludes. Work in progress extends the setting to a duopoly by studying how competitive forces shape the use of carbon offsets in consumer markets.

Related Literature: A large literature studies green consumerism in markets focusing mostly on how firms want to differentiate their products in respond to environmental preferences. Green consumerism has been studied in horizontal models of differentiation (Eriksson, 2004; Conrad, 2005) as well as in vertical models of differentiation. The first group of models of vertical differentiation study uni-dimensional differentiation along the environmental quality (Moraga-González and Padrón-Fumero, 2002; Lombardini-Riipinen, 2005; Deltas et al., 2013). A second group studies a bi-dimensional vertical differentiation based on hedonic and environmental quality by assuming perfect correlation between the two (Mantovani et al., 2016). Burani and Mantovani (2020) study a duopoly between sellers of a brown and a green product, who differentiate their product along the dimension of hedonic quality or environmental quality when firms account for consumer selection into the rival's product. Instead, I study a vertical differentiation model on the carbon offset dimension which can be sold to one consumer but mitigate the pollution emitted by the goods sold to other consumers.

Theoretically, it has been studied how carbon offsets affect donations in public good games (Vicary, 2000; Kotchen, 2009). However, they have not been studied as a screening tool by sellers in consumer markets.

2 Model Setup

I study a market served by a monopolistic firm offering a core product of normalized hedonic quality $q = 1$ to a continuum of consumers. Moreover, I normalize the marginal cost of the core product to 0. Each consumer has unit demand for the core product. The production of one unit of the core product generates a deterministic amount of carbon emissions $e > 0$, and the constant marginal cost of production is normalized to 0. The seller has access to reparation technology that allows bundling each unit of the core product with a quantity $o \geq 0$ of carbon offsets at a convex quadratic offsetting cost $C(o) = \frac{1}{2}\kappa o^2$, with $\kappa > 0$. I assume the seller takes κ as given and that the offsets reduce the aggregate externality generated by the production of the consumption goods.

I define a product-offset bundle as (q, o, p), where $q \in \{0, 1\}$ specifies whether the core product is included, $o \geq 0$ indicates the quantity of emissions offset, and $p \geq 0$ denotes the bundle price.[1] Consumers derive a gross valuation $v > 0$ from consuming the core product, but differ in their intrinsic motivation to reduce environmental pollution. A consumer's environmental type is indexed by a parameter $\theta \geq 0$. I assume that a fraction $\lambda \in (0, 1)$ of consumers are purely selfish ($\theta = 0$) and derive no disutility from environmental pollution. The remaining fraction $1-\lambda$ have environmental concerns and experience disutility from pollution. I assume that environmental types are distributed uniformly on $(0, \bar{\theta}]$. The seller cannot observe a consumer's environmental type θ, but knows the cumulative distribution function $F(\theta)$ and whether consumers care exclusively about the pollution of their own product or about total environmental pollution.

If consumers derive disutility exclusively from their own product's net pollution, the utility of type θ from purchasing bundle (q, o, p) is:

$$u(q, o, p; \theta) = qv - \theta \max\{qe - o, 0\} - p, \tag{1}$$

and the outside option utility from not purchasing any consumption-offset-bundle $(q = 0, o = 0, p = 0)$ is $u_0(\theta) = 0$ for all θ. Thus, consumers care about the carbon emissions associated with their consumption. Crucially, they do not experience any disutility from the negative externality generated by the products sold to other consumers.

In contrast, under outcome-based preferences consumers care about aggregate pollution in the market. Define $(q_\theta, o_\theta, p_\theta)$ as the product-offset-bundle the seller offers to consumer type θ. Then, $E = e \cdot \int_0^{\bar{\theta}} q_\theta \, dF(\theta)$ denotes aggregate emissions and $O = \int_0^{\bar{\theta}} o_\theta \, dF(\theta)$ aggregate offsets, so that $E_{\text{net}} = E - O$ represents aggregate net pollution generated in the market. Conditional on type θ purchasing bundle $c = (q, o, p)$, her beliefs regarding aggregate net emissions are given by the following:

[1] In principle, the seller is allowed to offer offsets to consumers without the core product $(q = 0, o > 0)$.

$$\mathbb{E}\left[E_{\text{net}} \mid c = (q, o, p)\right] = qe - o + \mathbb{E}\left[E_{\text{net}} \mid c = (0, 0, 0)\right], \tag{2}$$

where $\mathbb{E}\left[E_{\text{net}} | c = (0, 0, 0)\right]$ are her beliefs conditional on not purchasing the good. Her expected utility from purchasing bundle (q, o, p) is:

$$\mathbb{E}[u(q, o, p; \theta)] = qv - \theta\left(qe - o + \mathbb{E}\left[E_{\text{net}} \mid c = (0, 0, 0)\right]\right) - p. \tag{3}$$

If a consumer chooses not to purchase $(q = 0, o = 0, p = 0)$, they do not derive consumption utility but suffer a disutility from baseline aggregate net market pollution: $u_0(\theta) = -\theta\mathbb{E}\left[E_{\text{net}} \mid c = (0, 0, 0)\right]$. Thus, the net payoff from purchasing bundle (q, o, p) relative to the outside option is:

$$\mathbb{E}[u(q, o, p; \theta)] - u_0(\theta) = qv - p - \theta(qe - o). \tag{4}$$

When choosing to consume, the consumer does not evaluate the pollution generated by her consumption but internalizes the net pollution impact of their choice relative to non-purchase, $\theta(qe - o)$.

The seller offers $(q_\theta, o_\theta, p_\theta)$ to each consumer type θ. She chooses the optimal menu by maximizing expected profits subject to participation and incentive constraints. Moreover, I assume that the seller cannot offer more offsets than generated in the market. Thus, he solves the following maximization problem:

$$\max_{\{(q_\theta, o_\theta, p_\theta)\}_{\theta \in [0, \bar{\theta}]}} \int_0^{\bar{\theta}} \left[p_\theta - \frac{1}{2}\kappa o_\theta^2\right] dF(\theta) \tag{5}$$

subject to:

$$\text{(IC)} \quad u(q_\theta, o_\theta, p_\theta; \theta) \geq u(q_{\theta'}, o_{\theta'}, p_{\theta'}; \theta) \quad \forall \theta, \theta' \in [0, \bar{\theta}], \tag{IC}$$

$$\text{(IR)} \quad u(q_\theta, o_\theta, p_\theta; \theta) \geq u_0(\theta) \quad \forall \theta \in [0, \bar{\theta}], \tag{IR}$$

$$\text{(FEAS)} \quad \int_0^{\bar{\theta}} o_\theta \, dF(\theta) \leq \int_0^{\bar{\theta}} q_\theta e \, dF(\theta). \tag{FEAS}$$

The timing of the game is the following:

1. The seller posts a menu of product-offset options $\{(q_\theta, o_\theta, p_\theta)\}_{\theta \in [0, \bar{\theta}]}$.
2. Each consumer type θ privately select a product-offset bundle from the menu or choose their outside option.

3 Equilibrium Analysis

This section characterizes the resulting market equilibria under responsibility-based preferences in Section 3.1 and under outcome-based preferences in Section 3.2. All proofs are relegated to Appendix A.

3.1 Market Equilibrium under Responsibility-Based Preferences

To analyze the seller's optimal product design, I first examine consumer choice across available product-offset bundles. Selfish consumers ($\theta = 0$) do not value environmental quality and select the product bundle with the lowest posted price, purchasing if and only if $p \leq v$. Conversely, environmentally conscious types $\theta > 0$ evaluate the utility of purchasing a bundle (q, o, p) relative to their outside option of non-consumption. Under responsibility-based preferences, an environmentally conscious type purchases bundle $(1, o, p)$ if and only if:

$$v - \theta \max\{e - o, 0\} - p \geq 0. \tag{6}$$

I begin the equilibrium analysis by characterizing the optimal bundle offered to purely selfish consumers. Because selfish types derive no utility from voluntary carbon offsets, offering offsets to this segment increases the seller's production costs without extracting any additional surplus from selfish consumers.

Lemma 1 (Dirty Bundle for Selfish Consumers)**.** *Assume $\kappa > 0$ and $\lambda \in (0, 1)$. In any equilibrium, the seller offers the unoffset bundle $b_0^* = (q_0^* = 1, o_0^* = 0, p_0^* = v)$ to purely selfish consumers ($\theta = 0$).*

Lemma 1 establishes that the seller extracts the full surplus v from selfish consumers by selling the core product without carbon offsets. Crucially, offering an unoffset product to selfish consumers does not affect the demand of conscious types $\theta > 0$: any conscious type evaluating b_0^* obtains net utility $v - \theta e - v = -\theta e < 0$, and strictly prefers the outside option yielding zero utility.

Next, I characterize the optimal product menu offered to environmentally conscious types $\theta > 0$. Under responsibility-based preferences, conscious types care strictly about net emissions generated by their own product, and derive zero marginal utility from offsets exceeding their product's footprint ($o > e$).

Proposition 1 (Equilibrium under Responsibility-Based Preferences)**.** *Assume $\kappa > 0$ and $\lambda \in (0, 1)$. Under responsibility-based preferences, the equilibrium market outcome is characterized as follows:*

1. ***Coexistence of Clean and Dirty Products** ($v > \frac{1}{2}\kappa e^2$)**:** The seller offers a two-option menu consisting of:*

$$b_0^* = (1, 0, v) \quad \text{and} \quad b_{clean}^* = (1, e, v). \tag{7}$$

Selfish consumers ($\theta = 0$) purchase b_0^*, while all environmentally conscious consumers ($\theta \in (0, \bar{\theta}]$) purchase b_{clean}^*. Total seller profit is:

$$\Pi_{resp}^* = v - (1 - \lambda)\frac{1}{2}\kappa e^2. \tag{8}$$

2. ***Exclusion of Conscious Consumers** ($v \leq \frac{1}{2}\kappa e^2$)**:** The seller offers only $b_0^* = (1, 0, v)$. Selfish*

consumers purchase b_0^*, whereas conscious consumers choose non-consumption. Total seller profit is $\Pi^* = \lambda v$.

The economic intuition behind Proposition 1 rests on incentive compatibility. To prevent selfish consumers from selecting the bundle intended for conscious types $\theta > 0$, the price of the conscious bundle must satisfy $p_\theta \geq v$. However, a type-θ consumer's willingness to pay for a product with partial offsets ($e - o > 0$) is $v - \theta(e - o) < v$. Consequently, even if the seller extracts type θ's full willingness to pay for a partially offset product, the seller cannot prevent selfish consumers from purchasing that partially offset bundle. Together, the incentive compatibility constraint for selfish types and the participation constraints for conscious types force $o_\theta = e$ and $p_\theta = v$. Thus, partial offsets are incentive-incompatible, while overcompensation ($o > e$) is unprofitable. If offering full offsets is profitable ($v > \frac{1}{2}\kappa e^2$), the seller serves conscious types; otherwise, the seller excludes them from the market.

3.2 Market Equilibrium under Outcome-Based Preferences

I now characterize the market equilibrium when environmentally conscious consumers care about total pollution caused in the market. Recall from Section 2 that the net payoff of a type-θ consumer purchasing bundle $(1, o, p)$ relative to non-consumption $(q = 0, o = 0, p = 0)$ is:

$$\mathbb{E}[u(1, o, p; \theta)] - u_0(\theta) = v - p - \theta(e - o) = v - p + \theta(o - e). \tag{9}$$

Equation (9) highlights a key structural divergence from responsibility-based preferences: when the seller offers overcompensation ($o > e$), the term $\theta(o - e) > 0$ strictly increases a conscious consumer's willingness to pay above their hedonic product valuation v. Because outcome-based types care about total environmental damage, they are willing to pay to offset emissions generated by other market participants. Lemma 2 shows that the seller can extract this additional surplus from high environmental types.

Lemma 2 (Instability of Simple Product Dualism under Outcome-Based Preferences)**.** *Assume $\kappa e < \bar{\theta}$, $\lambda \in (0, 1)$, and $v > \frac{1}{2}\kappa e^2$. Under outcome-based preferences, simple product dualism $\{b_0^* = (1, 0, v), b_{clean}^* = (1, e, v)\}$ cannot be sustained as a market equilibrium.*

Lemma 2 demonstrates that offering only a standard fully offset product ($o = e$) leaves unexploited surplus on the table. Highly conscious types are willing to subsidize third-party emission reductions and, consequently, the seller has a profitable incentive to expand its product line by introducing overcompensating offsets ($o > e$). The following proposition characterizes the full profit-maximizing menu schedule.

Proposition 2 (Equilibrium under Outcome-Based Preferences)**.** *Assume $\kappa > 0$, $\lambda \in (0, 1)$, and $v > \frac{1}{2}\kappa e^2$. Under outcome-based preferences, let $\theta_c = \frac{\bar{\theta} + \kappa e}{2}$ define the over-remediation threshold under unconstrained demand. The profit-maximizing product menu schedule is characterized as follows:*

1. ***Dirty Option** ($\theta = 0$)**:** $b_0^* = (1, 0, v)$, selected by purely selfish consumers.*
2. ***Standard Clean Option** ($\theta \in (0, \theta_c]$)**:** $b_{clean}^* = (1, e, v)$, selected by low environmental types.*

3. ***Continuum of Super-Clean Options*** *($\theta \in (\theta_c, \bar{\theta}]$): High environmental types select custom over-compensated options $b^*(\theta) = (1, o^*(\theta), p^*(\theta))$.*

Market equilibrium outcomes depend on whether the aggregate offset feasibility constraint (FEAS) binds:

- ***Case 1 (No Over-Compensation,*** *$\bar{\theta} \leq \kappa e$): $\theta_c \geq \bar{\theta}$. The market reverts to standard product dualism with seller profit $\Pi^*_{outcome} = v - (1-\lambda)\frac{1}{2}\kappa e^2$.*
- ***Case 2 (Unconstrained Over-Compensation,*** *$\bar{\theta} > \kappa e$* ***and FEAS Unbinding):*** *When $(1-\lambda)\frac{(\bar{\theta}-\kappa e)^2}{4\kappa\bar{\theta}} \leq \lambda e$ holds, the seller sets $\mu = 0$, $o^*(\theta) = \frac{2\theta-\bar{\theta}}{\kappa} > e$, and total seller profit is:*

$$\Pi^*_{outcome} = v - (1-\lambda)\frac{1}{2}\kappa e^2 + (1-\lambda)\frac{(\bar{\theta}-\kappa e)^3}{12\kappa\bar{\theta}} > \Pi^*_{resp}. \tag{10}$$

- ***Case 3 (Capacity-Constrained Over-Compensation, FEAS Binding):*** *When $(1-\lambda)\frac{(\bar{\theta}-\kappa e)^2}{4\kappa\bar{\theta}} > \lambda e$, aggregate overcompensation is capped by total emissions from selfish consumers (λe). The seller faces shadow price $\mu^* = \bar{\theta} - \kappa e - 2\sqrt{\frac{\lambda}{1-\lambda}\kappa e\bar{\theta}} > 0$, over-remediation begins at $\theta^*_c = \bar{\theta} - \sqrt{\frac{\lambda}{1-\lambda}\kappa e\bar{\theta}}$, and $o^*(\theta) = e + \frac{2(\theta-\theta^*_c)}{\kappa}$.*

Proposition 2 reveals a novel economic mechanism: a profit-maximizing monopolist acts as an intermediary for third-party environmental externalities, cross-subsidizing offsets compensating the pollution emitted by selfish consumers with surplus extracted from high environmental types.

4 Welfare Implications and Policy Analysis

This section will discuss the social welfare implications of the consumer markets for the cases in which consumers care exclusively about the pollution emitted by their own product or the entire pollution in the market.

5 Conclusion

In this paper, I show that voluntary carbon offsets do help in mitigating pollution, however whether they are efficient depends on the nature of consumers' environmental preferences. If consumers care exclusively about their own carbon footprint and ignore the pollution caused by others, then market outcomes are inefficient. The pollution that is emitted by products sold to selfish consumers remains unaddressed because nobody feels responsible for them. A benevolent social planner acknowledges that these emissions need to be avoided and corrects the inefficiency with a carbon tax that is equal to the marginal social cost of pollution and thus forces the seller to internalize the aggregate pollution generated in the market. Moreover, this paper proposes theoretically a novel mechanism through which carbon offsets can operate in markets: a seller can offer them to consumers that are highly concerned about the environmental consequences of economic activity as a means to compensate for pollution generated by consumers that are not concerned about the environment.

In showing this mechanism, I made some simplifying assumptions. For example, I assumed that it can be fully verified whether the seller indeed carries out the offsets on behalf of consumers. It remains an open

question under which conditions on consumer preferences and sophistication this mechanism survives when the assumption of full verifiability is relaxed. Another assumption that I have made is that consumers are fully aware and fully informed about the pollution that is generated by their purchased product. However, it has been shown that there consumers systematically misperceive the carbon footprint of their purchased products (Imai et al., 2026). How the employment of voluntary carbon offsets interacts with the seller's commitment to an information structure that informs consumers about the pollution their product emits is a further interesting research avenue.

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A Omitted Proofs

A.1 Proof of Lemma 1

Let $b_0 = (q_0, o_0, p_0)$ denote the equilibrium bundle selected by selfish types ($\theta = 0$). Individual rationality (IR) for $\theta = 0$ requires $q_0 v - p_0 \geq 0$. Since production cost is normalized to zero, unit profit from selfish consumers is $\pi_0 = p_0 - \frac{1}{2}\kappa o_0^2 \leq q_0 v - \frac{1}{2}\kappa o_0^2$.

Suppose $o_0 > 0$, $p_0 < v$, or $q_0 = 0$. Consider a deviation where the seller replaces b_0 with $b_0^* = (1, 0, v)$. Type $\theta = 0$ obtains utility $v - v = 0$, satisfying IR. Any conscious type $\theta > 0$ evaluating b_0^* obtains net utility:

$$u(1, 0, v; \theta) = v - \theta e - v = -\theta e < 0, \tag{11}$$

which is strictly below their non-consumption utility $u_0(\theta) = 0$. Thus, no conscious type $\theta > 0$ selects b_0^*.

Per-capita profit on the mass λ of selfish consumers strictly increases to $v > p_0 - \frac{1}{2}\kappa o_0^2$, while the incentive constraints and choices of conscious consumers remain entirely unaffected. Hence, in any equilibrium, $q_0^* = 1$, $o_0^* = 0$, and $p_0^* = v$. ■

A.2 Proof of Proposition 1

By Lemma 1, the equilibrium bundle for selfish consumers is $b_0^* = (1, 0, v)$, yielding utility $u(1, 0, v; 0) = 0$ to selfish types.

To serve any conscious type $\theta > 0$ with a bundle $b_\theta = (q_\theta, o_\theta, p_\theta)$, two conditions must hold simultaneously:

1. **Incentive Compatibility for Selfish Types ($\mathbf{IC}_{0,\theta}$):** Selfish consumers must not prefer b_θ over b_0^*:

$$q_\theta v - p_\theta \leq 0 \implies p_\theta \geq q_\theta v. \tag{12}$$

2. **Individual Rationality for Type θ ($\mathbf{IR}_\theta$):** Conscious type θ requires non-negative net utility:

$$q_\theta v - \theta \max\{q_\theta e - o_\theta, 0\} - p_\theta \geq 0 \implies p_\theta \leq q_\theta v - \theta \max\{q_\theta e - o_\theta, 0\}. \tag{13}$$

Combining (12) and (13) yields:

$$q_\theta v \leq p_\theta \leq q_\theta v - \theta \max\{q_\theta e - o_\theta, 0\}. \tag{14}$$

For any $q_\theta = 1$ and $\theta > 0$, inequality (14) holds if and only if $\max\{e - o_\theta, 0\} = 0$, which implies $o_\theta \geq e$ and $p_\theta = v$. Thus, offering partial offsets ($o_\theta < e$) is impossible in equilibrium because setting $p_\theta < v$ causes selfish types to cannibalize b_θ, whereas setting $p_\theta \geq v$ violates IR for conscious types.

Because responsibility-based preferences imply zero marginal willingness to pay for offsets beyond product emissions ($o_\theta > e$), setting $o_\theta > e$ strictly increases offset cost $\frac{1}{2}\kappa o_\theta^2$ without allowing a price premium above $p_\theta = v$. Thus, the profit-maximizing offset level for conscious types is $o_\theta^* = e$.

Offering $b^*_{\text{clean}} = (1, e, v)$ yields unit profit $v - \frac{1}{2}\kappa e^2$ on the mass $1 - \lambda$ of conscious consumers. The seller includes b^*_{clean} in the menu if and only if $v - \frac{1}{2}\kappa e^2 > 0$.

If $v > \frac{1}{2}\kappa e^2$, total seller profit when serving all consumers is:

$$\Pi^*_{\text{resp}} = \lambda v + (1 - \lambda)\left(v - \frac{1}{2}\kappa e^2\right) = v - (1 - \lambda)\frac{1}{2}\kappa e^2. \tag{15}$$

If $v \leq \frac{1}{2}\kappa e^2$, offering b^*_{clean} is unprofitable, and the seller offers only $b^*_0 = (1, 0, v)$, yielding profit $\Pi^* = \lambda v$. ■

A.3 Proof of Lemma 2

Suppose $b^*_0 = (1, 0, v)$ and $b^*_{\text{clean}} = (1, e, v)$ constitute an equilibrium, yielding candidate profit $\Pi_{\text{cand}} = v - (1 - \lambda)\frac{1}{2}\kappa e^2$.

Consider a localized contract deviation where the seller retains b^*_0 and b^*_{clean}, but introduces an additional super-clean option $b_{\text{super}} = (1, o_\theta, p_\theta)$ for types above $\theta_c = \frac{\bar{\theta}+\kappa e}{2}$, where $o_\theta = e + \zeta$ and $p_\theta = v + \theta\zeta$ for a small $\zeta > 0$.

By construction, type θ obtains utility $v - p_\theta + \theta(o_\theta - e) = 0$, satisfying participation. The net change in unit profit from type θ relative to b^*_{clean} is:

$$\Delta\pi(\theta) = \left[v + \theta\zeta - \frac{1}{2}\kappa(e + \zeta)^2\right] - \left[v - \frac{1}{2}\kappa e^2\right] = \zeta\left(\theta - \kappa e\right) - \frac{1}{2}\kappa\zeta^2. \tag{16}$$

Taking the marginal profit derivative with respect to ζ at $\zeta = 0$ yields:

$$\left.\frac{\partial \Delta\pi(\theta)}{\partial \zeta}\right|_{\zeta=0} = \theta - \kappa e. \tag{17}$$

For all $\theta > \theta_c = \frac{\bar{\theta}+\kappa e}{2} > \kappa e$, this marginal profit is strictly positive. Thus, introducing super-clean options strictly increases firm profits, ruling out simple product dualism as an equilibrium. ■

A.4 Proof of Proposition 2

By Lemma 1, selfish consumers ($\theta = 0$, mass λ) receive $b^*_0 = (1, 0, v)$, yielding utility $U(0) = 0$ and producing uncompensated emissions λe.

For conscious consumers ($\theta \in (0, \bar{\theta}]$, mass $1 - \lambda$), indirect utility is $U(\theta) = v - p(\theta) + \theta(o(\theta) - e)$. By local incentive compatibility (Envelope Theorem), $U'(\theta) = o(\theta) - e$. Expressing price as a function of offsets and rent yields:

$$p(\theta) = v + \theta(o(\theta) - e) - U(\theta). \tag{18}$$

The seller's unit profit from type θ is:

$$\pi(\theta) = p(\theta) - \frac{1}{2}\kappa o(\theta)^2 = v - \frac{1}{2}\kappa o(\theta)^2 + \theta(o(\theta) - e) - U(\theta). \tag{19}$$

Integrating total information rents across conscious types $\theta \sim U(0, \bar{\theta}]$ via integration by parts yields:

$$\int_0^{\bar{\theta}} U(\theta) \frac{1-\lambda}{\bar{\theta}} d\theta = \frac{1-\lambda}{\bar{\theta}} \int_0^{\bar{\theta}} (\bar{\theta} - \theta)(o(\theta) - e) d\theta. \tag{20}$$

Substituting this into expected profit defines aggregate Virtual Surplus per conscious consumer:

$$\mathcal{J}(\theta, o) = v - \frac{1}{2}\kappa o^2 + (2\theta - \bar{\theta})(o - e). \tag{21}$$

To incorporate the aggregate feasibility constraint (FEAS) $\int_0^{\bar{\theta}} o(\theta) \frac{1-\lambda}{\bar{\theta}} d\theta \leq e$, we formulate the Lagrangian with multiplier $\mu \geq 0$:

$$\mathcal{L}(\{o(\theta)\}, \mu) = \lambda v + (1-\lambda) \int_0^{\bar{\theta}} \mathcal{J}(\theta, o(\theta)) \frac{d\theta}{\bar{\theta}} + \mu \left(e - (1-\lambda) \int_0^{\bar{\theta}} o(\theta) \frac{d\theta}{\bar{\theta}} \right). \tag{22}$$

Combining integrals yields:

$$\mathcal{L}(\{o(\theta)\}, \mu) = \lambda v + \mu e + (1-\lambda) \int_0^{\bar{\theta}} \left[v - \frac{1}{2}\kappa o(\theta)^2 + (2\theta - \bar{\theta})(o(\theta) - e) - \mu o(\theta) \right] \frac{d\theta}{\bar{\theta}}. \tag{23}$$

Differentiating pointwise with respect to $o(\theta)$ subject to $o(\theta) \geq e$ yields:

$$\frac{\partial \mathcal{L}}{\partial o(\theta)} = -\kappa o(\theta) + 2\theta - \bar{\theta} - \mu = 0 \implies o^*(\theta, \mu) = \max \left\{ e, \frac{2\theta - \bar{\theta} - \mu}{\kappa} \right\}. \tag{24}$$

Over-remediation ($o^*(\theta, \mu) > e$) begins at the threshold $\theta_c(\mu) = \frac{\bar{\theta} + \kappa e + \mu}{2}$. Aggregate over-compensation demanded by conscious types is:

$$O_{\text{extra}}(\mu) = (1-\lambda) \int_{\theta_c(\mu)}^{\bar{\theta}} \left(\frac{2\theta - \bar{\theta} - \mu}{\kappa} - e \right) \frac{d\theta}{\bar{\theta}} = (1-\lambda) \frac{(\bar{\theta} - \kappa e - \mu)^2}{4\kappa\bar{\theta}}. \tag{25}$$

By Karush-Kuhn-Tucker complementary slackness:

- **If $O_{\text{extra}}(0) \leq \lambda e$ (Case 1 and Case 2):** The feasibility constraint is non-binding ($\mu^* = 0$). Over-remediation threshold is $\theta_c = \frac{\bar{\theta} + \kappa e}{2}$, and $o^*(\theta) = \frac{2\theta - \bar{\theta}}{\kappa}$ for $\theta > \theta_c$.
- **If $O_{\text{extra}}(0) > \lambda e$ (Case 3):** Feasibility binds ($\mu^* > 0$). Setting $O_{\text{extra}}(\mu^*) = \lambda e$ yields:

$$(1-\lambda) \frac{(\bar{\theta} - \kappa e - \mu^*)^2}{4\kappa\bar{\theta}} = \lambda e \implies \mu^* = \bar{\theta} - \kappa e - 2\sqrt{\frac{\lambda}{1-\lambda} \kappa e \bar{\theta}} > 0. \tag{26}$$

  The constrained threshold is $\theta_c^* = \theta_c(\mu^*) = \bar{\theta} - \sqrt{\frac{\lambda}{1-\lambda} \kappa e \bar{\theta}}$, and $o^*(\theta) = e + \frac{2(\theta - \theta_c^*)}{\kappa}$ for $\theta > \theta_c^*$. ■